

Hayden Fuller
MCS HW/LAB 1
09/22/25

Theory Problems:

$$\tilde{s}(t) = \text{Re}\{s(t)e^{j\omega_c t}\}$$

$$\omega_c = 2\pi f_c$$

$$\text{BPSK } s(t) = A \cos(\omega_c t + a_i \pi)$$

$$\text{QPSK } s(t) = A \cos(\omega_c t + a_i \pi) - A \sin(\omega_c t + b_i \pi)$$

1) complex baseband representations

a) BPSK

$$a_i=0, s(t) = A \cos(\omega_c t)$$

$$a_i=1, s(t) = -A \cos(\omega_c t)$$

$$s(t) = A e^{j a_i \pi}$$

for $a_i=0,1$:

$$s(t) = A (-1)^{a_i}$$

b) QPSK

$$\text{similarly, } s(t) = A (-1)^{a_i} \cos(\omega_c t) - A (-1)^{b_i} \sin(\omega_c t)$$

$$s(t) = A((-1)^{a_i} + j(-1)^{b_i})$$

2) BER, theoretical expression for BPSK, QPSK

$$r(t) = s(t) + n(t)$$

$$n(t) = n_I(t) + j n_Q(t),$$

$$n_I(t), n_Q(t) \text{ i.i.d. } \sim N(0, \sigma^2), \sigma^2 = N_0/2.$$

a) conditional probabilities

$$p_{1|0} = P(A + n_I < 0) = P(n_I < -A) = Q(A/\sigma)$$

$$p_{0|1} = P(-A + n_I > 0) = P(n_I > A) = Q(A/\sigma)$$

$$p_{1|0} = p_{0|1} = Q(A/\sigma)$$

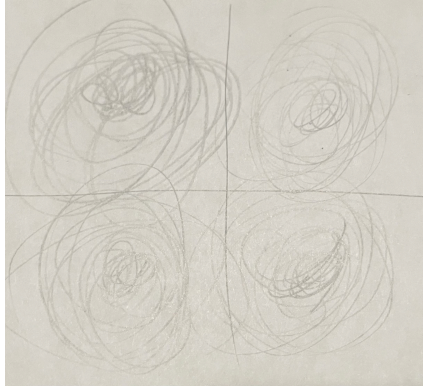
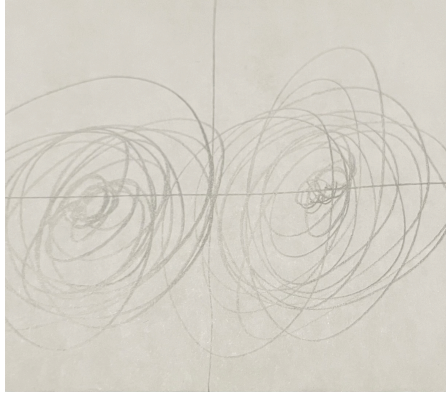
$$Q(A/\sigma) = \int_{A/\sigma}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

b) probability of error

$$p_e = .5p_{1|0} + .5p_{0|1} = Q(A/\sigma) = \int_{A/\sigma}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

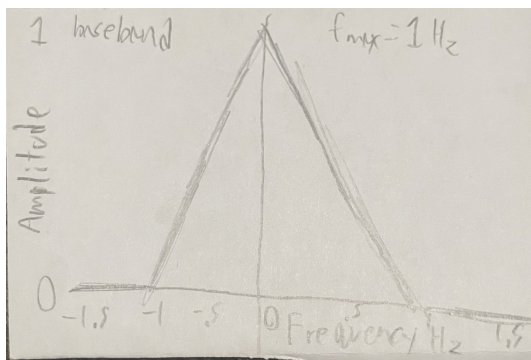
c) QPSK

QPSK is essentially two orthogonal BPSK streams (I and Q), each of which having their own gaussian noise and signal to noise ratio, giving identical p_e to BPSK. One sample of a QPSK streams has double the chance of error since it can be wrong in two different axis, but it carries twice as much information, so probability per bit stays the same.

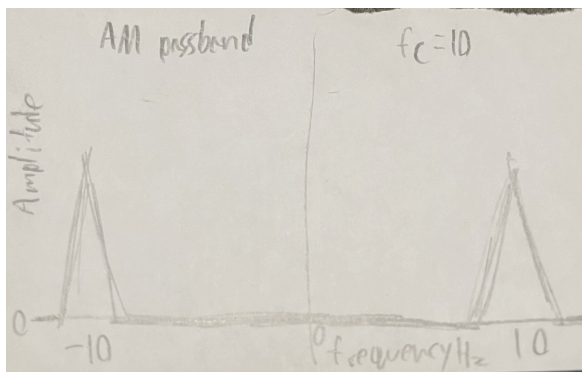


3) Spectrum of AM modulated signals

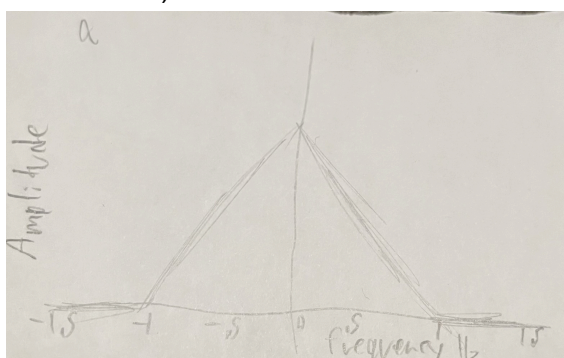
a) baseband



b) AM passband

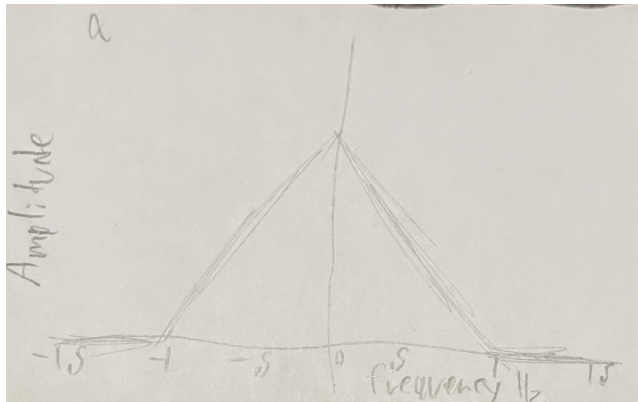


c) down converted

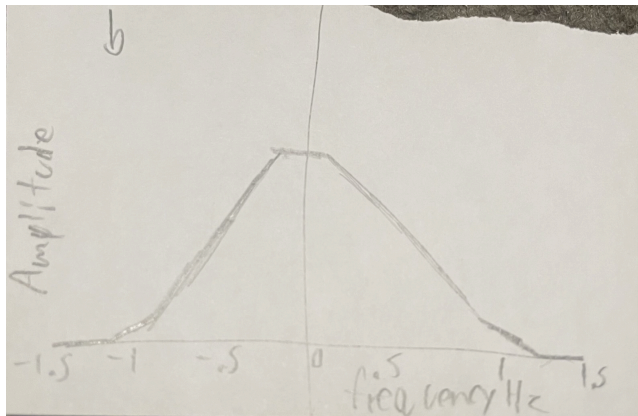


d) down conversion errors

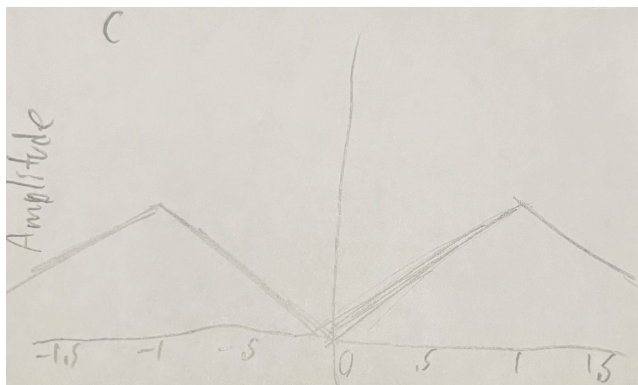
i) $x=0$



ii) $x=f_{\max}/(10f_c)$



iii) $x=f_{\max}/f_c$

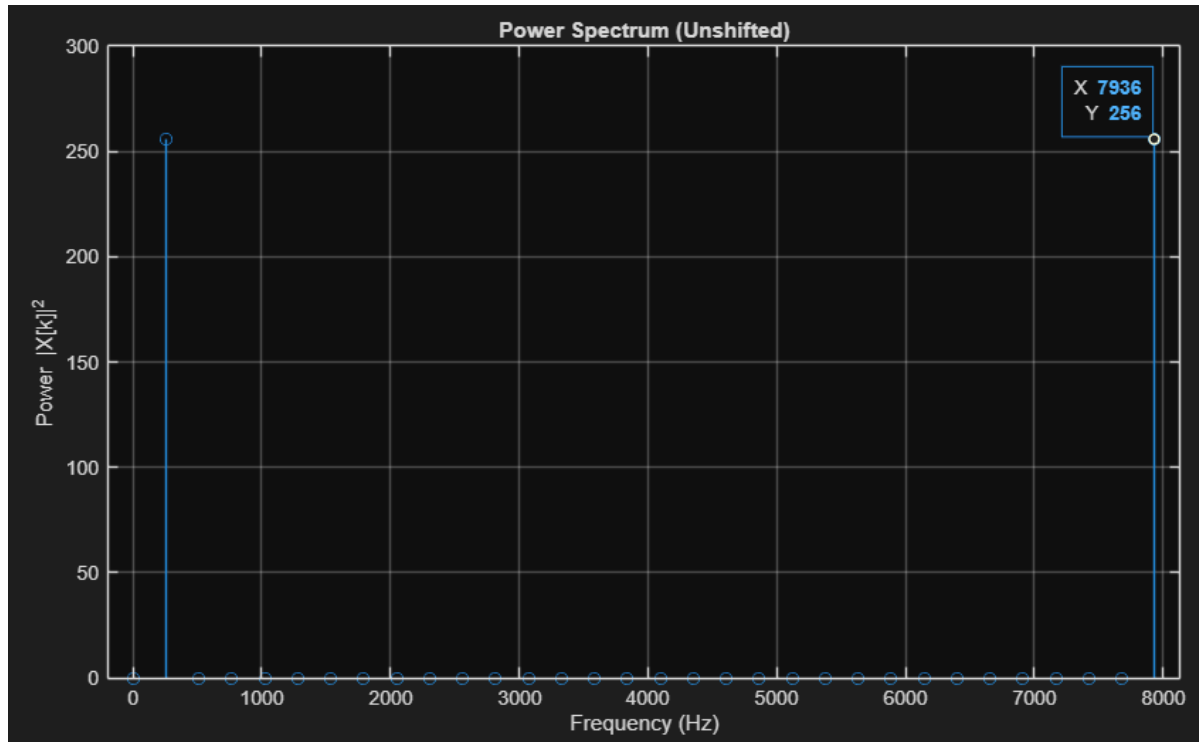


Matlab Simulations

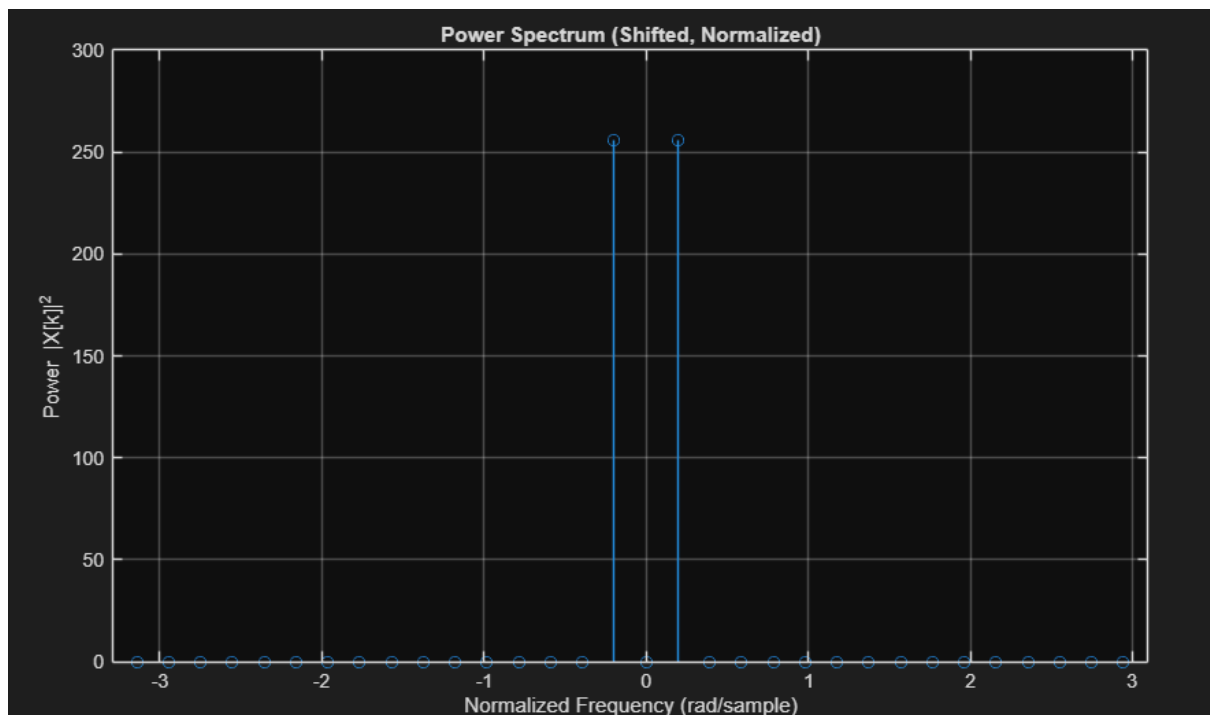
1) Spectral Analysis using the FFT

Deliverable: Include the plots generated from parts (c), (d), and (f). Note: Figures must include titles, labeled axes with units, and legends when relevant.

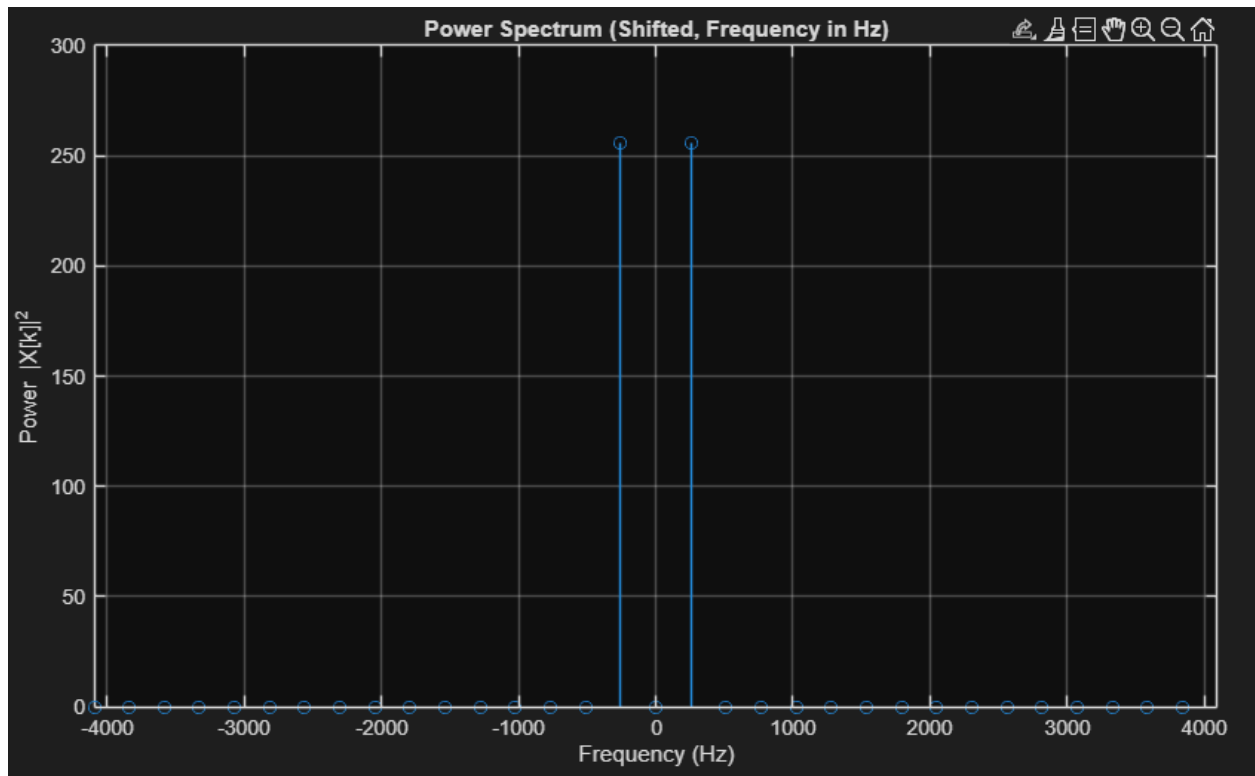
c



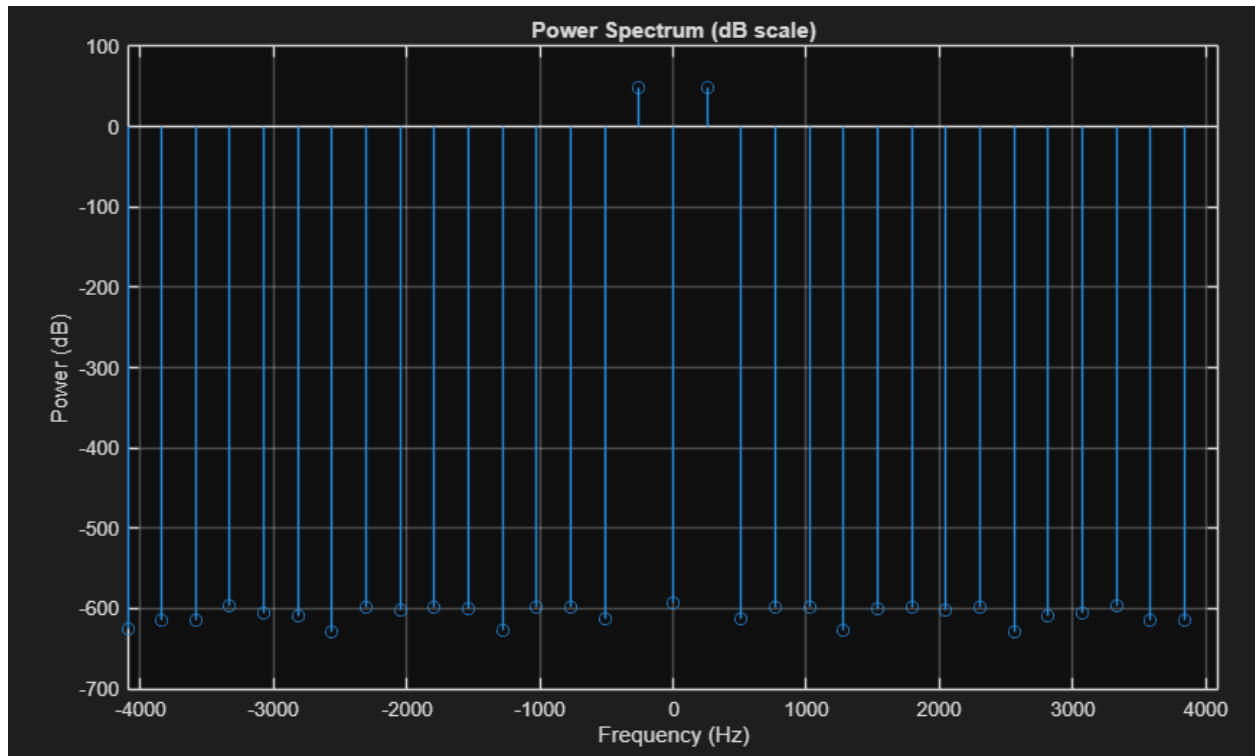
d



e



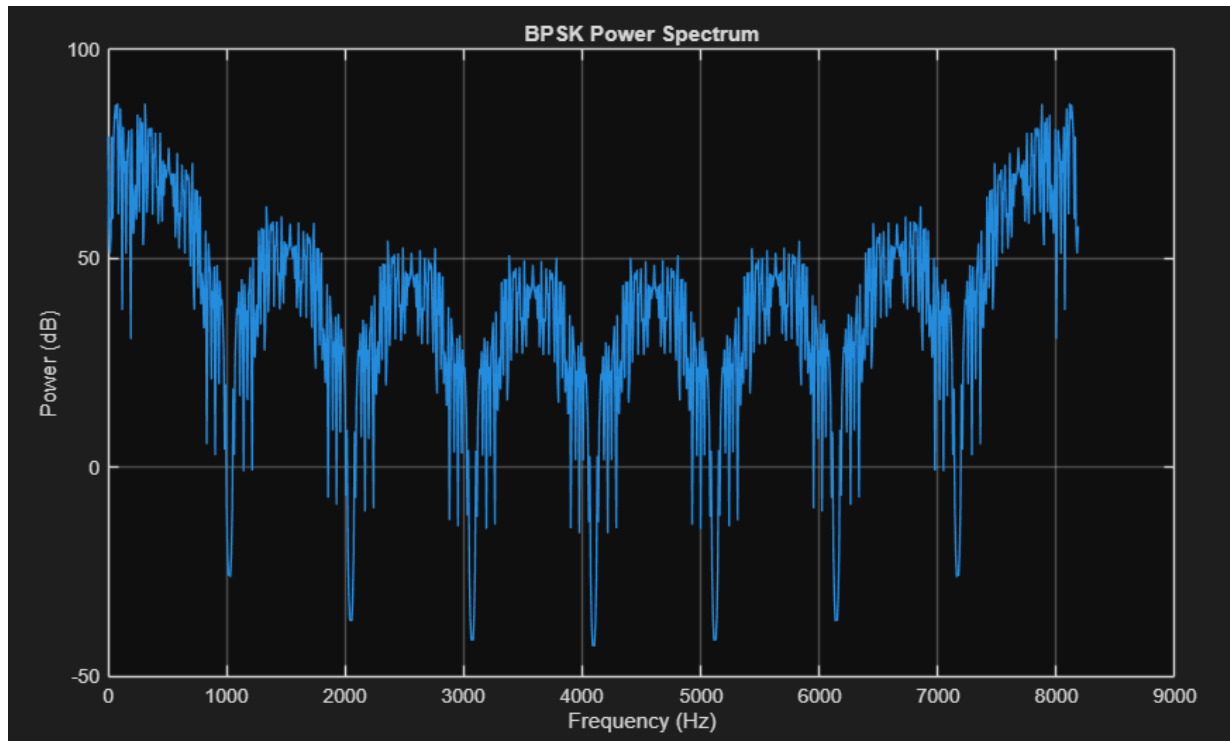
f



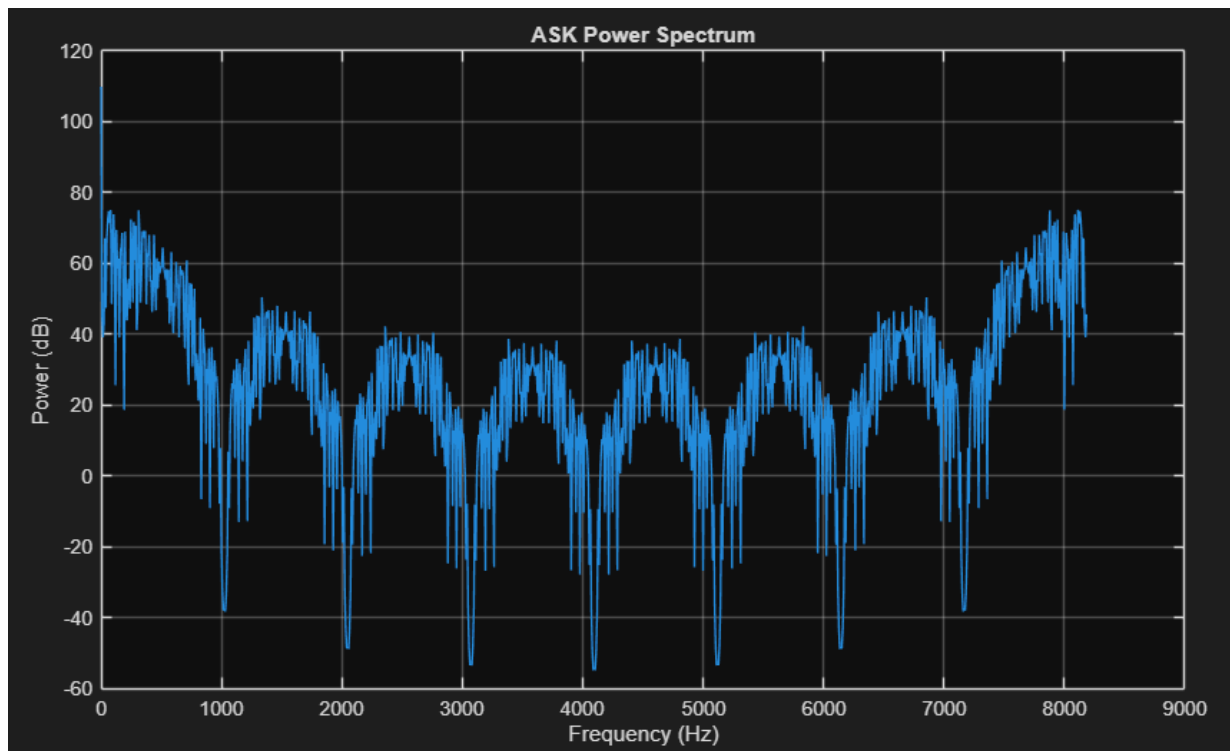
2) Power Spectrum

Deliverable: Include the three power spectra from parts (a)-(c). Note: Figures must include titles, labeled axes with units, and legends when relevant.

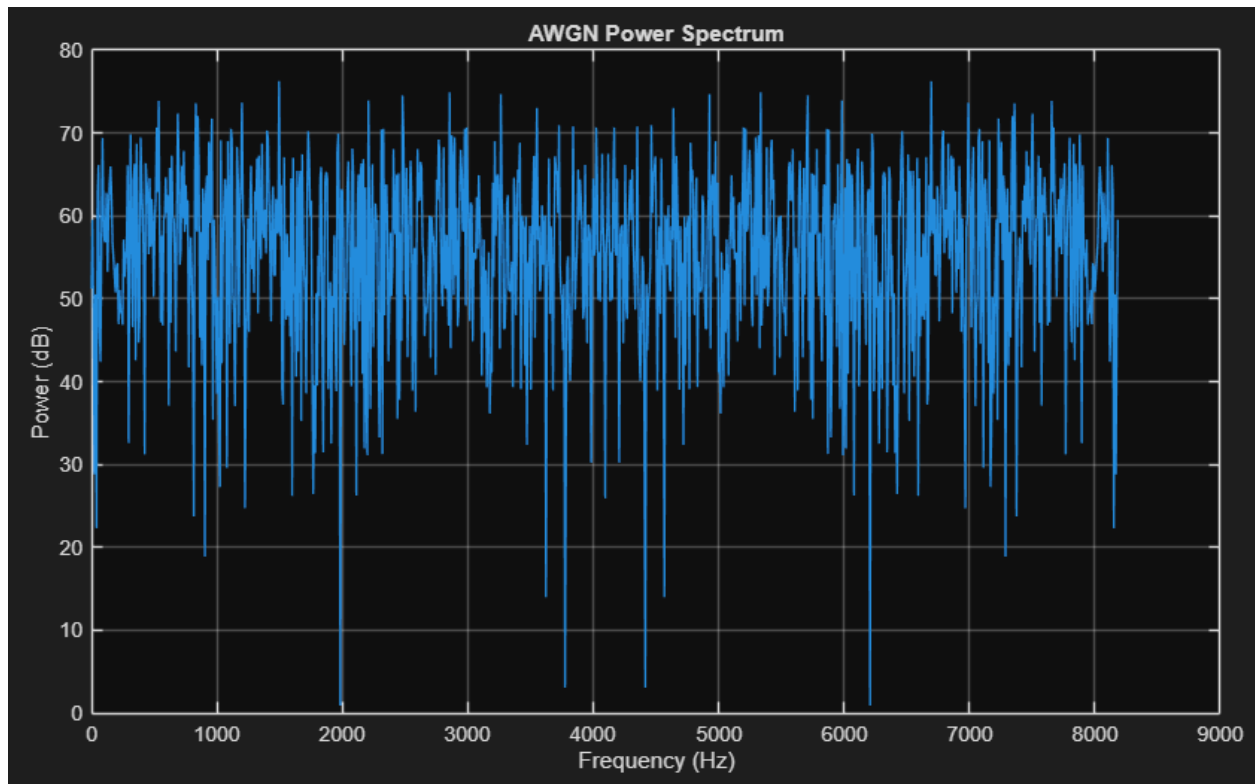
a



b



C



```

%% Hayden Fuller MCS HW1
clear
close all
fc = 256;
fs = 8192;
Nfft = 32;
A = 1;
Ts = 1/fs;
n = 0:Nfft-1;
x = A*cos(2*pi*fc*n*Ts);

%% a
X = fft(x, Nfft);

%% b
P = abs(X).^2;

%% c
freqs_Hz_unshifted = (0:Nfft-1)*(fs/Nfft);
figure;
stem(freqs_Hz_unshifted, P)
title('Power Spectrum (Unshifted)')
xlabel('Frequency (Hz)')
ylabel('Power |X[k]|^2')
grid on

%% d
P_shifted = fftshift(P);
f_norm_shifted = (-Nfft/2:Nfft/2-1)/Nfft * 2*pi;
figure;
stem(f_norm_shifted, P_shifted)
title('Power Spectrum (Shifted, Normalized)')
xlabel('Normalized Frequency (rad/sample)')
ylabel('Power |X[k]|^2')
grid on;

%% e
freqs_Hz_shifted = (-Nfft/2:Nfft/2-1)*(fs/Nfft);
figure;
stem(freqs_Hz_shifted, P_shifted)
title('Power Spectrum (Shifted, Frequency in Hz)')
xlabel('Frequency (Hz)')
ylabel('Power |X[k]|^2')
xlim([-fs/2 fs/2])
grid on;

%% f
figure;
stem(freqs_Hz_shifted, db(P_shifted))
title('Power Spectrum (dB scale)')
xlabel('Frequency (Hz)')
ylabel('Power (dB)')
xlim([-fs/2 fs/2])
grid on;

```

```

fc = 256;
fs = 8192;
Nfft = 1024;
A = 1;
Ts = 1/fs;
n = 0:Nfft-1;
x = A*cos(2*pi*fc*n*Ts);

%% a
X = fft(x, Nfft);

%% b
P = abs(X).^2;

%% c
freqs_Hz_unshifted = (0:Nfft-1)*(fs/Nfft);
figure;
plot(freqs_Hz_unshifted, P)
title('Power Spectrum (Unshifted)')
xlabel('Frequency (Hz)')
ylabel('Power |X[k]|^2')
grid on
;
%% d
P_shifted = fftshift(P);
f_norm_shifted = (-Nfft/2:Nfft/2-1)/Nfft * 2*pi;
figure;
plot(f_norm_shifted, P_shifted)
title('Power Spectrum (Shifted, Normalized)')
xlabel('Normalized Frequency (rad/sample)')
ylabel('Power |X[k]|^2')
grid on;

%% e
freqs_Hz_shifted = (-Nfft/2:Nfft/2-1)*(fs/Nfft);
figure;
plot(freqs_Hz_shifted, P_shifted)
title('Power Spectrum (Shifted, Frequency in Hz)')
xlabel('Frequency (Hz)')
ylabel('Power |X[k]|^2')
xlim([-fs/2 fs/2])
grid on;

%% f
figure;
plot(freqs_Hz_shifted, db(P_shifted))
title('Power Spectrum (dB scale)')
xlabel('Frequency (Hz)')
ylabel('Power (dB)')
xlim([-fs/2 fs/2])
grid on;

```

```

%% Part 2
Nfft = 1024;
%%fs = 1;
freqs = (0:Nfft-1)*(fs/Nfft);
fc = 256;
fs = 8192;
%% Nfft = 32;
A = 1;
Ts = 1/fs;

%% a BPSK
x_bpsk = randsrc(1, 128);
x_bpsk = kron(x_bpsk, ones(1, 8));
X_bpsk = fft(x_bpsk, Nfft);
P_bpsk = abs(X_bpsk).^2;
P_bpsk_dB = db(P_bpsk);

figure;
plot(freqs, P_bpsk_dB)
title('BPSK Power Spectrum')
xlabel('Frequency (Hz)')
ylabel('Power (dB)')
grid on;

%% b ASK
x_ask = (x_bpsk + 1) / 2;
X_ask = fft(x_ask, Nfft);
P_ask = abs(X_ask).^2;
P_ask_dB = db(P_ask);

figure;
plot(freqs, P_ask_dB)
title('ASK Power Spectrum')
xlabel('Frequency (Hz)')
ylabel('Power (dB)')
grid on;

%% c AWGN
x_awgn = randn(1, Nfft);
X_awgn = fft(x_awgn, Nfft);
P_awgn = abs(X_awgn).^2;
P_awgn_dB = db(P_awgn);

figure;
plot(freqs, P_awgn_dB)
title('AWGN Power Spectrum')
xlabel('Frequency (Hz)')
ylabel('Power (dB)')
grid on;

```